ajar ARENA JOURNAL OF ARCHITECTURAL RESEARCH

Liekens J, et al. Political Dimensions of Research by Design. *ARENA Journal of Architectural Research*. 2022; 7(1): 4. DOI: https://doi.org/10.5334/ajar.334

DESIGN RESEARCH ESSAY

Political Dimensions of Research by Design

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In this overview essay, which also introduces the other contributions to the *AJAR* Special Collection about the Erasmus+-funded research project on 'Mapping, Reflecting and Developing PhD-by-Design Programs', we as the guest editors venture from a situated position based on our experiences of that collaborative process. From that stance, and as part of the dynamics sprouting from the Mapping/Reflecting/Developing project, we focus here on one specific characteristic of research-by-design – that of the inherently political dimension in positioning such research. It is a dimension which often seems to be ignored or under-articulated within discussions about Research-by-Design projects and PhD-by-Design programs. To counteract that deficiency, the notion of the 'political' is connected in this introductory essay to the realms of architecture, architectural practice, and architectural design research, on multiple levels. For the other essays which make up this Special Collection, this essay presents what we call the 'agonistic table'.

Keywords: Research-by-design; PhD-by-Design; architectural design methods; methodological assemblages; research communities

The situated background

Many schools of architecture and urban design across Europe have during recent decades initiated and implemented new PhD programs. In addition, to challenge the implementation of curricula that simply import existing formats from other disciplines with strong research traditions, such as the natural or social sciences – where research problems located at the *core* of architecture and design practice remain largely unaddressed – many schools have likewise taken initiatives to develop design-based education at doctoral level.

Within the context of a fast-growing body of design-based PhD education, the background to this *AJAR* Special Collection came from an Erasmus+ Strategic Partnership project that ran from 2018–21. Under the title of 'Mapping, Reflecting and Developing PhD-by-Design Programs', its intention was firstly to map the various approaches to design-based architectural research and design-based doctorates which have emerged during recent decades, and secondly to relate, position and develop further the different streams of carrying out research-by-design. The overall goal was neither to create a comprehensive and full analysis of this extensive landscape, nor to rank or evaluate the various ways of doing this kind of research, but rather to explore the rich range of approaches – and, in doing so, to contribute to reflexive and creative positionings, implementations, and further developments of new and existing doctoral programs. As many European schools of architecture are currently developing PhD programmes, there is a strong need to offer overviews of the concepts, methods, and curricula within research-based education, and to be able – by analysing existing concepts, methodological tools and teaching modules – to assemble, develop, and adapt to the specific resources, cultures, and institutional histories of various schools of architecture.

The Erasmus+-funded project was formed by four European universities with different traditions and experiences of architectural practice, education and research. Two of the participating universities have long traditions and established PhD programmes in architecture which include Research by Design methodologies (KU Leuven's Campus Sint-Lucas in Ghent/Brussels in Belgium, and Chalmers University of Technology

in Gothenburg, Sweden), while two are in the stage of establishing design-based doctoral programs (University of Liechtenstein in Vaduz, and Bergen School of Architecture in Norway). Organised around three work streams – i.e. the mapping/reflecting/developing of relevant concepts, methods, and curricula – the research project was constructed as a series of meetings, workshops, round-table discussions and collaboratively produced outputs (**Figures 1–3**). With participants coming from different environments and contexts, the intention was to stage dialogues that could integrate their varying perspectives in experience-based exchanges of ideas, knowledge, and competences.



Figure 1: Discussion on mappings and research design with participants from the four partner universities at the first meeting of the 'Mapping/Reflecting/Developing' project, held at the University of Liechtenstein in Vaduz in November 2018 [Photo courtesy of the authors].

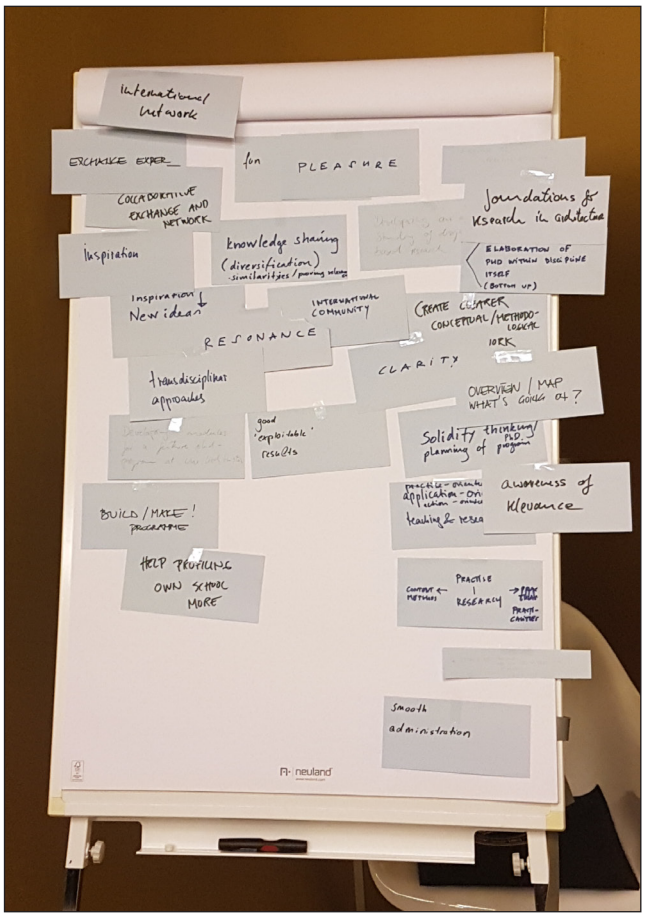


Figure 2: One of the first unelaborated mappings of concepts and directions of the research project coming out of the initial discussions [Photo courtesy of the authors].

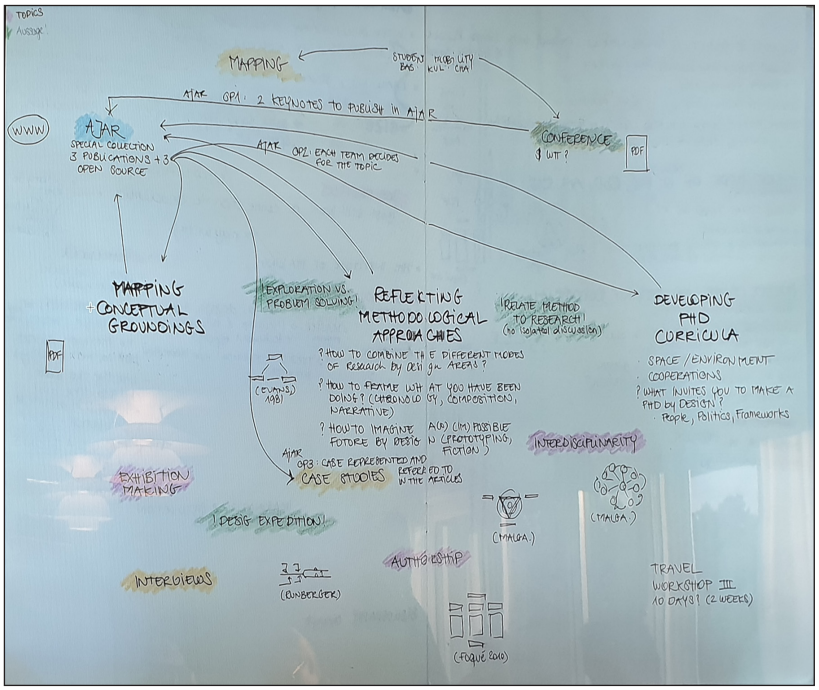


Figure 3: Intermediate mapping of the project and its relations of contents and works, produced at a workshop session with the project participants [Courtesy of the authors].

Each contribution to this *AJAR* Special Collection hence aims to provide insights into the conceptual underpinnings, methodological tools and curricular cultures which can support research-based education in architectural schools. This introductory essay offers a background frame and briefly introduces the respective essays, while also attempting to add to the debate. When we as guest-editors came together to decide how to introduce this Special Collection, two observations occurred to us. Firstly, we realised that each of the essays makes its propositions from a particular *situated* position. As such, the authors speak from within a specific research milieu, program, curriculum, school, institution, when they are discussing the shared terrain of research-by-design. Secondly, there is a reoccurring presence of an inherently political dimension in the contributions, something which would seem to be characteristic for research-by-design, but which we also had the sensation was somewhat overlooked or rendered absent in the essays.

Mirroring the contributions, we decided that we had to base our introductory essay on a similarly situated position, including aspects we thought specifically interesting to foreground in terms of the political dimension of design-based architectural research. All three of us guest editors had different roles and perspectives – as PhD candidate, PhD assessor, and PhD supervisor – yet had collaborated with each other in a 'double-degree' doctorate by Johan Liekens which was carried out between KU Leuven's Faculty of Architecture (Campus Sint-Lucas) and the Department of Architecture and Civil Engineering at Chalmers University of Technology [1]. The critical, political, and ethical capacities of architectural artifacts were investigated as part of that PhD project, and the methodological discussion in the thesis argued that an inherently political dimension was characteristic for research-by-design. Hence, in this introductory essay we will expand upon the notion of how the 'political' colours design-driven research. But first we will briefly introduce the other contributions.

Essay contributions

This Special Collection includes essays by participants from the 'Mapping/Reflecting/Developing' project and by invited keynote speakers at the concluding conference: Jonathan Hill from the Bartlett School of Architecture, University College London, and Gro Rødne from the Faculty of Architecture and Design at the Norwegian University of Science and Technology.

In his essay on 'Design Research: The Next 500 Years', Jonathan Hill presents and situates the Bartlett's pioneering Architectural Design doctoral programme, the very first PhD-by-Design program in the world and in which he has been involved in and directed from its very beginning. Hill emphasizes that even though many PhD-by-Design programs define themselves as being different from traditional doctorates, the Bartlett has always seen theirs within the mould of what a typical PhD is expected to be like. Consciously named

the PhD Architectural Design to avoid any misconception of theory just being text-based, the significant difference is that the Bartlett PhDs combine a design project and a text that share a research subject in a productive relationship. Hill notes, as he has done in other writings, that to treat design research as if it is new to architecture – something that contemporary discourse often does – is to ignore the history of the architect since the Italian Renaissance, and more specifically the methods and means which have been invaluable to architects for over 500 years since. Architects write about themselves, their work, and their projects, and indeed the most influential architects have always tended to write, draw and publish as well as build. Knowledge production in architecture thus involved relationships between drawings, texts and buildings in multi-directional ways, forming interdependent webs of influences that together have contributed to architects' development over the centuries. Hill argues that in the formation of modern identity in more recent centuries, with diary-writing and other processes of self-examination being introduced, a sense of objectivity is an increasing aspiration, yet no diary is ever entirely truthful, and no writer fails to edit, reinvent, and alter the past while reflecting upon it, which also influences the future. Like a written history, any design project is a reinterpretation of the past in the present, and one which, equivalent to fiction, convinces viewers/users to suspend disbelief. Even if Hill acknowledges that there is no exact formula for an architectural design doctorate, he underlines the contemporary relevance of interdisciplinary research and notes that many of Bartlett's PhD Architectural Design theses deal with socially engaged work in actual 'real-world' sites. This makes the graduates particularly conscious of issues of architectural authorship, using the academic context to reflect upon their roles in practice situations, and whereby the built environment is conceived as an educational resource through which marginalised voices can be key in a more effective, situated production of knowledge.

Gro Rødne's keynote talk was the basis for the essay on 'Making is Thinking: From Design Fixation to Provocative Competence', written together with Leif Martin Hokstad. Their point of departure is in the broader perspective of architectural education, with the observation that students today seem reluctant to move out of their comfort zone, instead requesting easier and more direct routes to learning. Simultaneously, there is a growing need within contemporary society to deal with challenges across disciplines, often characterized by ill-defined 'wicked' problems. This then requires abilities and mindsets which can deal with situations of uncertainty and ambiguity. To meet this challenge, Rødne and Hokstad present the theoretical framework of what they term 'Making is Thinking', an approach they have developed in their teaching practice at Trondheim which suggests potential pathways to a revised pedagogical stance which aims towards transformative learning. Building upon Shkolvsky's theories of strange-making or de-familiarization, and Barrett's pedagogy of strangeness and uncertainty, the intention of their 'Making is Thinking' approach is to enhance counter-intuitive design and make exercises more troublesome, while simultaneously constructing a learning environment that balances risk with trust, confidence and belonging. Within this context, they suggest the notion of 'provocative competence' as an aspect of skills which can facilitate innovation, and offer a way to develop discrepancies and dissonances that trigger a move away from habitual positions and repetitive patterns. When practicing their methods, they also include 'real-world' urban interventions and public events involving actual stakeholders. The approach opens itself up to the strange and unpredictable, allowing for intuition and for risk, with Rødne and Hokstad arguing that mistakes and failures are integral parts of a creative process.

Also focussing on an educational perspective, but at third level of research-based education, Charlotte Erckrath, Vera Kaps and Cecilie Andersson in their essay, 'Curricular Culture in PhD by Design: Perspectives Gained from the Aarhus School of Architecture', investigate design doctorates in European architectural schools. Various programs have evolved to include research-by-design (RbD) at doctoral level as a means to establish a school's identity and thus to help student recruitment. In this context, the RbD approach seems to have arisen more naturally as a strong vehicle for motivation in some architecture schools, whereas others have had to make more efforts to complement and adjust their existing programs. Based on their study of PhD programs, Erckrath, Kaps and Andersson present an idiosyncratic map with the intention of revealing key stakeholders and players, alongside a chronology in which various European networking activities have happened. Here they note that the Aarhus School of Architecture in Denmark has been particularly involved and visible in discussions about the role of design research in architecture. They then take this as the basis for a more in-depth, case-study analysis of the Aarhus School. The authors argue for the strengthening of a curricular culture for design-based PhD programs, pointing out potential strategies and components while at the same time highlighting challenges that may occur. Although they note that RbD is practiced at the Aarhus School of Architecture as a research method without a clearly framed methodology, leading consequently to several confusions and insecurities, they nonetheless show that situatedness, a unique mindset,

and specific competences have created a curricular culture that is grounded in the Aarhus School's general design and research cultures. Therefore, it can be seen to be a conscious means to shape a school's identity, promote student recruitment, and raise the profile to increase competitiveness.

Two other essays in the Special Collection, written by Krystyna Pietrzyk and by Malgorzata Zboinska, focus particularly on the methodological aspects of Research by Design as analysed by various theorists [2; 3]. Pietrzyk starts off her essay on 'Wicked Problems in Architectural Research: The Role of Research by Design' from her experience as the director of a Graduate School of Architecture. She notes that PhD students often have difficulties in referring to relevant theories and systems of inquiry when they are designing the research set-up for their theses. She observes that research questions and hypotheses are often formulated to address 'wicked problems', which then causes methodological challenges, not least for more inexperienced researchers. To be able to devise and use a proper methodology, a suitable ontological and epistemological perspective must be chosen, positioning the researcher and the specific project consciously, which therefore calls for an awareness of the wider epistemological landscape. Situating the discussion in relation to traditional and established research traditions, as well as to the theory of 'wicked problems', Pietrzyk argues for the potential role of research-by-design as an important methodology to investigate and resolve these 'wicked problems'. Since the latter often come from architectural practice, they are thus expected to be solved by practice, with solutions not being true or false but rather good or bad according to applied norms – in contrast to issues in basic sciences which generally are resolved within a theoretical space under ideal conditions. Instead, when addressing complex societal challenges, such as issues of sustainability or heritage, an architectural research project often needs to consider many aspects ranging from physical, ecological, technological, socio-political through to economic ones and others related to values and norms. Looking at two examples of PhD theses, Pietrzyk argues that research-by-design can be seen as an approach to address 'wicked problems' in architectural design or urban planning, in which the mode of inquiry includes insights into future conditions which are uncertain and with consequences that are difficult to predict.

Malgorzata Zboinska, in her contribution titled 'Architectural Research in Hybrid Mode: Combining Diverse Methods within Design-Based Architectural Research Inquiry', devises a framework for architectural research as a 'hybrid method' which she argues covers a different typology of approaches beyond the traditional qualitative/quantitative distinction. Based upon analysis of a range of design-based doctoral dissertations from seven internationally renowned institutions, she proposes a synthetic distinction between three main categories of design research – artistically driven, architecture-centred, and science-inspired – which she then uses as a foundation to help underpin the notion of 'hybrid method' of design research in architecture, as a kind of complement to the often-referred concept of 'mixed method' research. When discussing the three different categories, she points to challenges and weaknesses in each, with for instance more artistically driven research methods potentially leading to non-compliance with the classical markers of academic research, such as reproducibility of experiments or publication in scientific journals, thus causing researchers to have to find other ways to ensure proper recognition of their work – a task especially hard and burdensome for junior artistically-minded researchers. Zboinska argues that her version of the 'hybrid method' has its paramount strength in the rich milieu it offers for knowledge production and communication, yet also notes that the diversity of perspectives, operational modes, ways of forming/exploring research questions, methods of data gathering, knowledge carriers, and dissemination media all form a complex methodological milieu that requires advanced levels of scholarship and experience of numerous different research approaches.

The final essay in this Special Collection is 'Dissecting the Archipelago: PhD by Design Concepts in the Fields of Architecture and Urban Design', and is written by Hanne Van Reusel, Christoph Michels and Yves Schoonjans. In their essay they try to map the constantly growing discourse about, and conceptual frameworks for, designerly ways of conducting design-based research. Van Reusel, Michels and Schoonjans suggest viewing the numerous conceptual contributions as an 'emerging archipelago' of different attitudes and positions that PhD candidates, supervisors, researchers – and indeed PhD programs – need to navigate, find their position within, and contribute to. They present this research landscape as a constructed, artificial archipelago, constantly subject to modifications, new imaginings, and, not least, political struggles over its future. To cut through this archipelago, they present five cross-sections on the topics of science/design, subjectivity, disciplinarity, literacies and practice/theory. Van Reusel, Michels and Schoonjans acknowledge they cannot know all the motivations, desires and conditions that drive design-based PhD students, or the institutions they are part of, and thus their mapping might not help all readers. However, they believe that a positioning in relation to these five cross-sections is a relevant exercise for any PhD-by-Design project. They state that it is less important where on (or off) the map a PhD project or program happens to be located, and

their aim is not not to strengthen specific positions but to encourage PhD candidates to make explicit their own positions to make their research more discussable.

Political Dimensions of Research by Design

While these essay summaries convey some of the contents of the larger research project, they do not speak of the actual debates, tensions, and struggles of strategic and political dimensions that occurred along way. In saying this, we do not mean that the research project was especially difficult or troublesome, but rather that the work involved mirrored the wider heated debates in the field of Research by Design. Even though it is possible to sense some of these issues in the essays in this Special Collection, we will now discuss in more detail one specific dimension – namely the political. It is something that has been part of discussions about research-by-design, and thus it also inescapably coloured the activities of our 'Mapping/Reflecting/Developing' project.

As noted, the political dimension was a key aspect of the previously mentioned doctoral thesis by Johan Liekens. An ambition of that PhD project was to argue for a more conscious deployment of architectural design as productively conflictual, in terms of being dis-sensual (rather than merely consensual) and agonistic (**Figure 4**). The specific dimension of the methodology used by Liekens only became clearly formulated after an exchange of ideas and positions with the appointed future opponent, who, within the Scandinavian PhD context, is the jury member that at the public defence frames the work and then critically interrogates it. Here it is worth noting the quite remarkable terms used in academic structures, such as 'defence', 'opponent', 'critical interrogation', and so on, which in themselves radiate a sense of political battles playing

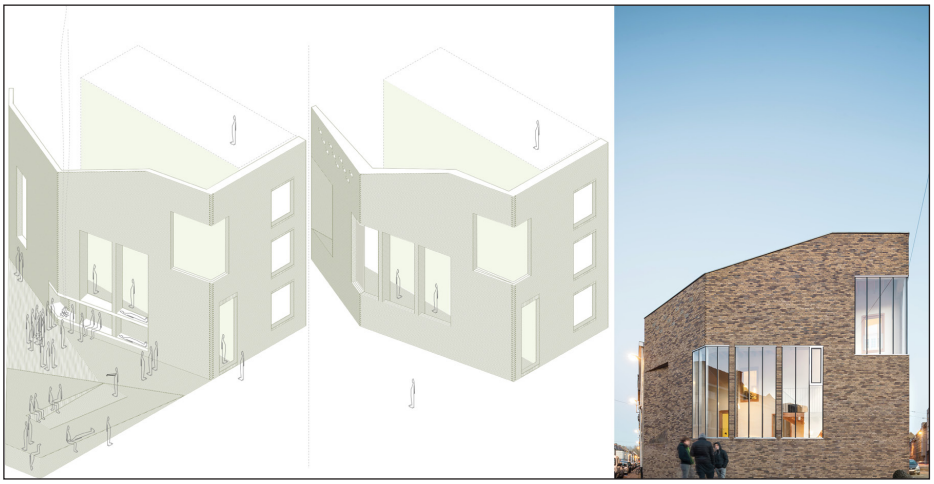


Figure 4: Architecture, architectural practice and architectural design research as agonistic stages. Drawing and image of the 'Walled House' project, excerpted from the PhD project by Johan Liekens [Drawing by Johan Liekens/STUDIOLO architectuur; photograph by Stijn Bollaert].



Figure 5: Round-table debates arranged at KU Leuven Faculty of Architecture Campus Sint-Lucas in Gent in May 2019 on two topics: 'The personal, the inter-subjective and the common in Research by Design' and 'How to teach research-by-design in architecture in a PhD program' [Courtesy of the authors].



Figure 6: Round-table debates arranged at KU Leuven Faculty of Architecture Campus Sint-Lucas in Gent in May 2019, on the same two topics [Photograph by Christoph Michels].

out between positions and territories. The dynamics of a PhD project leading up to the defence can thus easily be described as political in themselves, as indeed can also the trajectory of the 'Mapping/Reflecting/Developing' project. This then suggests an institutional/political dimension to the academic status of research-by-design, meaning that the latter is not yet a self-evident terrain or a methodology that is already fully stable and established.

When stating that the political characteristic immediately relates to the subject matter of architecture, it should be noted that this idea has typically been only hesitantly or even reluctantly received within the architectural realm. This is well expressed throughout Pier Vittorio Aureli's lecture on 'Can Architecture be Political?' [4]. Herein, Aureli advances what he considers a paradox. At one end of the paradox, he resolutely holds that architecture cannot be political. On the other, architecture appears to be inevitably political. According to Aureli, it cannot be truly political mainly because architectural practices have been historically developed and anchored in ideologies of *consensus*. This has even been strengthened, he argues, by a strict division of labour in which all actors involved in architectural practices and processes of making and experiencing – from designer to maker to occupant – are concerned only with specific, isolated parts of those processes. At the other end of the paradox, holding that architecture is in fact always political, Aureli proposes the idea that architecture is political even down to the level of its smallest and most intimate details. Architectural elements and forms always address a particular socio-spatial condition, something which by itself implies an idea of the political. Any idea of space as such implies an idea of the political, and any idea of the political implies an idea of space. Architecture as a practice of framing spaces in that sense always addresses an idea of the political. In his effort to solve the paradox, Aureli deploys a somewhat uneasy distinction between architecture as a *form* – an agentic substance, matter, element, artefact – and architecture as a *practice*. He states that while architectural form is or can be political, architectural practice cannot. We find it difficult to agree with such a distinction, and would rather argue for architectural practice, as well as the specific architectural practice that is research-by-design, as indeed being fundamentally political.

There are obvious examples and accounts of architectural form – substance, matter, element, artefact – being political. One may think of architectural elements mediating inside and outside, cis- and trans-, above and below, belonging to or being pushed out, and enabling certain activities and relations while obstructing other. In contemporary architectural research, there have also been formulated more complex and intriguing accounts of how the political comes into play. These accounts touch on architectural form, but also on design research, or research-by-design, as a mode of practice. In this context, one can refer for instance to Albena Yaneva and Chantal Mouffe as providing two analyses which are of relevance for the mapping, reflecting and developing of research-by-design.

In exploring the field of architecture from the perspective of sociology, Albena Yaneva identifies architectural artefacts not as consensual conglomerates, but, conversely, as congregations of controversy [5]. It has led her to develop a particular ethnographic research method with which to *follow* architecture 'in-flight', thereby mapping the 'controversies' of the genesis and life of architectural artefacts. In resonance with Yaneva, but in a way swapping Yaneva's observational or even retrospective method with a more active, future-orientated approach, the political philosopher Chantal Mouffe argues that aesthetic practices such as architecture should deploy themselves more consciously as *agonistic* practices, i.e. as argumentative, polemical, combative exchanges. Such practices can, according to Mouffe, help to produce new realities through the production of what she calls democratic, 'agonistic public space' [6: p. 119]. Her concept of *agonism* here refers to a divided common ground in which there is 'an agreement on shared principles, but a disagreement on the interpretation of these principles'. The agonistic stage, including architecture and architectural practice, is a process of 'conflictual consensus' where 'oppositional positions can be played out' [6: p. 92, 102]. Mouffe's idea can be paired with a vision about, and a distinction between, two kinds of pluralism. One typically conceives of pluralism in the sense of the circulation of different points of view, interest, and value. It states that, while one is unable to embrace them all, they nevertheless constitute a harmonious ensemble. But another view on pluralism, which is the one that Mouffe clings to, is rooted in the certainty that 'all these different and multiple views cannot be reconciled', thus implying an inherent dimension of antagonism. The idea of a harmonious ensemble is considered nonsensical, with Mouffe emphasizing that:

> *… accepting the fact and existence of pluralism implies … accepting the fact of antagonism, of conflict. Conflict that is ineradicable, that cannot be reconciled. In fact, that is exactly what I understand as antagonism.* [6: p. 124–125].

We therefore want to introduce the ideas of Yaneva and Mouffe to link the political to architecture and architectural practices, such as design research, exactly because of their resonance with how we often perceive the congregation of voices whenever research-by-design is being negotiated. In our opinion, these are – indeed, necessarily are – agonistic voices, constructed on a common but divided ground, and sharing principles but disagreeing productively about the interpretation of those principles. The stages in which that happens, such as the 'Mapping/Reflecting/Developing' project, should thus be openly seen as agonistic, public, and pluralistic construction sites within the ongoing knowledge project of research-by-design.

An obvious political trait affecting architecture, and the specific architectural practice of research-by-design, is suggested also by Herbert Simon in his seminal book, *The Science of Artificial*. He notes that while scientific disciplines focus upon understanding 'how things are', design-based practice and related activities aim more at 'how things ought to be' [7: p. 129]. The same goes for architectural practice, claims Kim Dovey, arguing that it needs to point out directions, to enable and constrain, and thereby it assumes responsibility. Dovey goes as far as to define architecture along this line of reasoning, declaring that 'in order to be classed as "architecture" there must be some vision for the future … at stake' [8: p. 257–258]. Aligning with these perspectives, Alain Findeli identifies (architectural) designers as wanting to 'modify human-environment interactions and to transform them into preferred ones'. Hence, they approach reality in a 'prescriptive and diagnostic' mode rather than in merely descriptive ways. According to Findeli, this approach also fundamentally affects research-by-design. 'Endowed with the intellectual culture of design', these design researchers – who according to Findeli cannot be anyone else except trained designers – 'look for what is going wrong in the world', and so have an orientation towards improvement and transformation. In responding to diagnosis and moving towards prescription, the epistemological stance of the design researcher is therefore 'projective' [9: p. 293]. Through their projects – whether a spatial and artefactual entity, or a mode of practice – architects (and architectural design researchers) conceive alternative realities as a commitment to the future, aspiring to at least some degree of change. This interpretation of the future is again political. Does one want to propose projects as routes towards making the world better? Or does one aim at making the discipline better, such as in improving the aesthetic quality of artefacts and experiences? Or, perhaps more opportunistically, does one see the strengthening of the reputation of a designer or an institution as a way to compete more successfully for research funding or academic reputation? In each case, or even if a mixture of them all, a position needs to be taken, and taking that position is a political act.

In accepting these political aspects, architectural design research is also opened up beyond the confinement and autonomy of what some claim to be its own terrains. If architecture, and design research, involves itself in diagnosing what is going wrong, prescribing methods to tackle the problems, and projecting alternatives futures, then in our opinion it cannot but intermingle with other disciplines. In this transdisciplinary

enterprise of knowledge building, design research has a particular potential and quality in its capacity to 'complement the scientific analysis because design is trained in combining issues of facts with issues of values' [10: p. 81]. This development of a sense of responsibility and response-ability among architects about things going wrong often applies to concerns that are more widely shared in society [11; 12: p. 130]. When also there are different interpretations of what it means to deploy this diagnostic, prescriptive and projective stance, a further political aspect of research-by-design is unleashed. It has been present at the various venues that we as design-driven researchers have engaged in, plus it could be seen in the debates and dynamics of the 'Mapping/Reflecting/Developing' project. These differences along a spectrum of positions can at times be productive, whereas at other times they become more problematic. While some voices at the table promote research-by-design along transdisciplinary lines, using design intelligence 'from within' to work on problems that transcend the discipline or profession, other contrary voices argue passionately for a more autonomous position. The latter believe that the inquiry should take place in their own terrain, on their own terms, and through their own selected (and thus also arguably restricted) modes, methods and media. While this is perhaps an understandable feeling in a research area that is still developing, the autonomous stance is in our opinion too defensive and potentially hampers the reach which design research could potentially have. It needs to be a matter of balance, of negotiating how one's own terrain should be blended with others, and when that blending process becomes a diluting one. Closely affiliated with this stance of working 'from within' to transcend beyond, Catharina Dyrssen points out that the focus in much of Nordic design research culture increasingly shifts away 'from reflections on the artists' creative processes to a renewed interest in art-related materiality and agency', resulting in research themes that are 'driven by complex art-society issues related to capacities, possibilities and agency of artistic production and modes of investigation', and also to ways of not only using 'theory projected from other domains … but to develop theory and conceptualisation through artistic approaches and practices' [13: p. 176].



Figure 7: Workshop discussion with projects participants in one of the transnational meetings, here at Chalmers University of Technology in Gothenburg in November 2019 [Courtesy of the authors].



Figure 8: Round-table debate and presentations by invited researchers and practitioners on the topic of methodological approaches, arranged at Chalmers University in November 2019 [Courtesy of the authors].

It is to be noted that in the above discussion there is a tension between architectural practice as such – as in for example the crafting of architectural artefacts – and design research, or research-by-design, to which we already referred as constituting a specific architectural practice. That there is an intimate relationship between both aspects is certain. What both arguably share is the acknowledgment of the disciplinary specificity of design as a practice and mode of research, as has been pointed out by authors such as Jane Rendell and Nigel Cross [14; 15]. The role of design within design research is thus, according to Henk Borgdorff, that of a 'methodological vehicle' [16: p. 46]. In the view of Chris Rust and others, design becomes an instrument for 'research in which the professional and/or creative practices of art, design and architecture play an instrumental part in an inquiry' [17: p. 11]. A clear political position to be negotiated for each design research is to us evident. Does design and practice lead, ground, base or serve the research investigations, or, conversely, does design simply equal research? According to Linda Groat and David Wang, if one looks at design/practice on one hand, and at research on the other, then both 'embody many important similarities, including many complementary and overlapping qualities.' However, they argue that design/practice and research constitute 'relatively distinct kinds of activity', each with its own unique qualities [2: p. 23]. Peter Downton thus describes design as 'a way of inquiring, a way of producing knowing and knowledge'. Therefore, he argues, design is a way of researching. According to Downton, much work still needs to be done in constructing arguments for this seemingly self-evident proposition – an effort in which a central role is played by debates about knowledge and ways of knowing [18: p. 2].

In the specific milieu of the Sint-Lucas School of Architecture – which nowadays is the KU Leuven Faculty of Architecture, Campus Sint-Lucas – this political dimension is strong. It has also been translated into a variety of support structures set up for former research training sessions as well as the current doctoral training process. A spectrum of conceptualizations exists, ranging from design research been seen as almost equalling design, to design research envisaging design activity more as a methodological vehicle, though to design research not needing to contain any design activity whatsoever. Some of these conceptualizations call for a radical break with the more established, traditional ways of conducting academic research, while others absolutely count upon the latter. What is needed again is a conscious balancing. As Christopher Frayling and others indicate, the extreme positions miss the point that there is in fact 'a continuum from scientific research to creative practice' [19: p. 15]. Within Frayling's continuum, each researcher is invited to choose a position, negotiate a hybrid of methodologies to deploy, and in doing so define their ideas about the role of design. Such positioning does not only relate to research methods; instead, it is a positioning in relation to other kinds of research, and to the people conducting it, often developing their research into new realms. By imagining these adjacent research worlds connecting, and from our experiences in the 'Mapping/Reflecting/Developing' project, we could see the formation of schools of research, and communities of thought, within design research – not merely coinciding with institutional groupings but with researchers' shared perspectives as a community based upon proximity [20: p. 5]. To form such communities is and always should be a matter of negotiation and politics, and not the imposition of any singular, emblematic model of design research.

Research methodology is often in focus in debates about design research, and many times it is connected to the importance of positioning oneself as a researcher within a continuum, or else in opposition to specific positions, thereby negotiating a personal as well as communal mixture of methods. Such negotiation is of course not just a matter of selecting from an arsenal of established methods. Rather the opposite, research methods also need to be invented or crafted, not least in mapping, reflecting and developing PhD-by-Design programmes. The notion of research method, along with its relation to the production of knowledge, is likewise permeated by politics, as has been suggested by those such as John Law, who argues that research methods are not solely methods but in fact reflect one's being and aspirations. They point to how the world is seen, or to how one hopes to remake the world [21: p. 10], this latter reflection aligning with the prescriptive or normative stance referred to above. To use a research method or to assemble, gather and create different methods into the methodological apparatus of design research is hence a fundamentally political deed. Law states that research methods re-craft 'new versions of the world' from a 'hinterland of realities'. As such, any research method 'makes new manifestations and new concealments, and it does so continuously' [21: p. 141]. Beyond just being procedural applications for understanding reality, they are performative, active agents in creating new realities, and new versions of the world. Realities, according to Law, need to be 'practiced' [21: p. 15], and it is in this practicing of realities that research methods will have their way. They gather or arrange things; they conceal parts of reality or reveal latent aspects of it. The same obviously goes for the gathering of methods into a multiple, mixed or hybrid methodological approach, something that is often seen as characteristic for research-by-design. Selecting or creating methods into these kinds of gathering is also to obscure or repel other methods which have not been chosen. Thus, as Law emphasizes, every such

act of arranging as in a method or a gathering of methods has 'political implications' [21: p. 141]. He calls on us to recalibrate our understanding of research methods accordingly: 'We will need to teach ourselves to know some of the realities of the world using methods unusual or unknown' [21: p. 2]. This means to consider research methods anew, to craft methods, to imagine them, to invent them. Law says that we can gain knowledge through 'embodiment' as well as through a range of activities, and he specifically mentions architecture as a method that can help to address the full and complex 'textures' of reality [21: p. 146]. This highlights once more an affinity between architecture and the practice of research-by-design.

From the perspective of research-by-design, and how research methods function therein, and moving closer to the dynamics of manifesting and concealing, it is notable that the political philosopher Jacques Rancière describes a fundamental role to be played by aesthetic practices such as architecture. Aesthetic practices, Rancière suggests, are inherently political, just as much as political practices are inherently aesthetic. Architecture as an aesthetic practice has the eminent capacity of rendering matters present, while in the same movement and moment rendering others absent. In this way it can introduce disputes about, and subsequently 're-partition', the sensible [22: p. 37], resulting in a shifting of that which can be sensed by people, and which makes sense within a particular terrain of reality. The architectural theorist Roemer Van Toorn also characterizes architecture as political in this sense, 'in the manner in which it makes reality visible by means of its own aesthetic syntax … giving it direction' [23: p. 60]. Close to this, the philosopher William McNeill speaks of the creation of 'worlds' and 'gatherings' in which certain possibilities are opened up, while others never occur to us; he describes this as an active and political process which both selects and leaves out, thereby assembling new gatherings [24]. Architecture, and hence design research, can serve in assigning and holding things, matters, perspectives in their proper places – a 'policing activity', as Rancière would argue – but it can just as much be deployed as a 'political activity', inflicting a rupture in that which makes sense, making sense anew, and inviting the proper and improper to swap place or shift position [22; 25]. As a 'policing activity', architectural practice can avoid conflicts from being played out, or even take place. As a 'political activity', in contrast, it can invite negotiations to be staged, and from those it can create new articulations of reality. Hence architecture and design research operate as reconfiguring practices of sense-making, and along with that, as transformative practices of reality-making and world-making.

The assembling, reviewing, reworking and editing carried out by the 'Mapping/Reflecting/Developing' project, thus makes sense in a rich terrain which is still – at least partially – open for exploration, configuration and reconfiguration. It helps to discover or make worlds within already existing worlds, exploring new assemblages and enabling new articulations to make sense. In his book on *Ways of Worldmaking*, Nelson Goodman proposes that there is not one stable world which exists as a unified and consensual entity, but rather 'a diversity of right and even conflicting versions or worlds in the making' – and thus there are 'multiple actual worlds coexisting' that are 'of independent interest and importance, not reducible to a single base' [26: p. 2–4]. Negotiations between these co-existing worlds is a propelling dynamic, orientated to the process of



Figure 9: Round-table gatherings and other phenomena around the topic of 'Complicating Machines', deploying themselves as agonistic stages for negotiation and sense-making in the research process for the PhD thesis by Johan Liekens [Photographic collage by Johan Liekens].

finding common ground. The image of world-making as painted by Goodman also introduces an inherent political dimension. Rancière too speaks of the 'configuration of a specific world' as being part of politics, and he contends that 'there is politics because there are competing worlds' [25: p. 7]. We suggest that we should also conceive of research-by-design, and the venues in which it is debated, and being part of the political making of worlds. We should likewise think of research methods as involved in the making of worlds.

Concluding remarks

The staging of the efforts to map, reflect and develop research-by-design in a non-homogeneous congregation of institutions involving a wide range of people, such as the 'Mapping/Reflecting/Developing' project brought around the table, is a matter of politics in many of the senses discussed above. To refer to Chantal Mouffe's terminology, around this table we are multiple, sharing the table as an agonistic space in which 'oppositional positions can be played out' [6: p. 92]. The table is hence a space of 'conflictual consensus', and of a divided common, raising political questions for the authors in this Special Collection. What is research-by-design, what role does design have in it, when did design research emerge, and how and to what extent does it relate to or differ from scientific or artistic research? What should research-by-design be aimed at? How are theory and practice related in design research? Which are the literacies required for research-by-design? How can one conceive of the personal and the collective within design research, and in the production of knowledge? What borders does design research assume, and what borders does it need to transgress? These are just a few of many possible political questions. On these questions there is indeed agreement about shared principles, but there may well be variety in, and even disagreement about, the interpretation of those principles.

Thinking about the contribution that these essays in this Special Collection make to the field of research-by-design, and borrowing words from Groat and Wang, they can best be considered as 'increments of knowledge attained through a variety of means' [2: p. 8]. Each author speaks from their situated positions, as well as within the congregation of partners that came together at various stages in the 'Mapping/Reflecting/Developing' project, and by doing so they add to our existing registers in an accumulative way, thereby changing those registers and furthering knowledge. One might again refer to Goodman's idea that 'worlds are as much "made" as that they are found and to know a world one must make a world ... comprehension and creation happen together' [26: p. 22]. This has exactly been the aim of the Mapping/Reflecting/Developing project, and it has been substantiated by the authors of the contributions in this Special Collection. These different voices invite everyone to further debates and discussions around tables about the subject of design research.

Funding Information

This essay, as well as the others included in this *AJAR* Special Collection, are part of the Erasmus+ research project titled 'Mapping, Reflecting and Developing PhD-by-Design Programs', grant number 2018-1-LI01-KA203-000108.

Competing Interests

The authors have no competing interests to declare.

References

1. **Liekens J.** *Architecture's Poetic Instrumentality: Developing the Critical, Political, and Ethical Capacities of Architectural Artifacts.* PhD dissertation. KU Leuven Faculty of Architecture, Campus Sint-Lucas & Chalmers University of Technology, Department of Architecture and Civil Engineering; 2020.
2. **Groat LN, Wang D.** *Architectural Research Methods.* John Wiley; 2013.
3. **Fallman D.** The Interaction Design Research Triangle of Design Practice, Design Studies, and Design Exploration. In *Design Issues*, 24(3). MIT Press; 2008: 4–10. DOI: https://doi.org/10.1162/desi.2008.24.3.4
4. **Aureli PV.** Can Architecture be Political? In: The Architecture Exchange Series #2, Talk #3. Architecture Exchange; 2014.
5. **Yaneva A.** *Mapping Controversies in Architecture.* Routledge; 2012.
6. **Mouffe C.** Democracy Revisited (in conversation with Chantal Mouffe). In: Miessen M (ed.). *The Nightmare of Participation.* Sternberg Press; 2010.
7. **Simon HA.** *The Sciences of the Artificial.* MIT Press; 1981.
8. **Dovey K.** I Mean to Be Critical, But …. In: Rendell J et al (eds). *Critical Architecture.* Routledge; 2007: 252–260.

9. **Findeli A.** Searching for Design Research Questions: Some Conceptual Clarifications. In Chow R, Jonas W, Gesche J (eds). *Questions, Hypotheses & Conjectures: Discussions on Projects by Early Stage and Senior Research Designers*. Universe; 2010: 286–303.
10. **Janssens N.** *Utopia-Driven Projective Research: A Design Approach to Explore the Theory and Practice of Meta-Urbanism.* PhD dissertation. Chalmers University of Technology, Department of Architecture and Civil Engineering; 2012.
11. **Barad K.** *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning.* Duke University Press; 2007. DOI: https://doi.org/10.2307/j.ctv12101zq
12. **Haraway D.** *Staying with the Trouble: Making Kin in the Chthulucene.* Duke University Press; 2016. DOI: https://doi.org/10.2307/j.ctv11cw25q
13. **Dyrssen C.** Precision: The Compositional Accuracy of Artistic Judgement. In: Nilsson F, Dunin-Woyseth H, Janssens N (eds). *Perspectives on Research Assessment in Architecture, Music and the Arts: Discussing Doctorateness.* Routledge; 2017: 175–189. DOI: https://doi.org/10.4324/9781315526652-12
14. **Rendell J.** Architectural Research and Disciplinarity. In *Architectural Research Quarterly*. 2004; 8(2): 141–147. DOI: https://doi.org/10.1017/S135913550400017X
15. **Cross N.** *Designerly Ways of Knowing.* Springer-Verlag; 2006.
16. **Borgdorff H.** The Production of Knowledge in Artistic Research. In: Biggs M, Karlsson H (eds). *The Routledge Companion to Research in the Arts.* Routledge; 2010: 44–63.
17. **Rust C, Mottram J, Till J.** *Review of Practice-led Research in Art, Design & Architecture.* Arts and Humanities Research Council; 2007.
18. **Downton P.** *Design Research.* RMIT University Press; 2003.
19. **Frayling C,** et al. *Practice-based Doctorates in the Creative and Performing Arts and Design.* UK Council for Graduate Education; 1997.
20. **Dunin-Woyseth H, Nilsson F.** Building a Culture of Doctoral Scholarship in Architecture and Design: A Belgian-Scandinavian Case. In: *Nordic Journal of Architectural Research.* 2011; 23(1): 41–55.
21. **Law J.** *After Method: Mess in Social Science Research.* Routledge; 2004.
22. **Rancière J.** *Dissensus: On Politics and Aesthetics.* Continuum; 2010. DOI: https://doi.org/10.5040/9781472547378.ch-001
23. **Van Toorn R.** Aesthetics as a Form of Politics. In: Seijdel J, Melis L (eds). *Open! Art, Culture & the Public Domain.* NAI 010 Publishers; 2006: 54–61.
24. **McNeill W.** Design and the Enigma of the World. In: Willis AM (ed.). *Design Philosophy Papers.* 2006; (2). DOI: https://doi.org/10.2752/144871306X13966268131550
25. **Rancière J.** The Thinking of Dissensus: Politics and Aesthetics. In: Bowman P, Stamp R (eds). *Reading Rancière.* Continuum; 2011: 1–17. DOI: https://doi.org/10.5040/9781472547378.ch-001
26. **Goodman N.** *Ways of Worldmaking.* Hackett Publishing Company; 1978.

How to cite this article: Liekens J, Nilsson F and Janssens N. Political Dimensions of Research by Design. *ARENA Journal of Architectural Research*. 2022; 7(1): 4. DOI: https://doi.org/10.5334/ajar.334

Published: 11 March 2022

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